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EDITORIAL

It is with great pleasure and a deep sense of academic fulfillment that we present the maiden edition of the Public Sector Accounting and Finance Journal (PSAFJ), published by ANAN University, Kwall. This inaugural issue marks a significant milestone in the University's commitment to advancing scholarship, policy, and practice in the field of public sector accounting, finance, and governance.

In an era when nations are striving for transparency, accountability, and fiscal sustainability, the role of public sector accounting and finance has never been more critical. The journal is therefore conceived as a platform for intellectual engagement, rigorous research, and policy discourse that contribute meaningfully to the strengthening of public financial management systems, good governance, and sustainable development in Nigeria and beyond.

This maiden edition features ten scholarly articles drawn from a wide spectrum of research interests and methodologies within the field. The papers collectively address critical issues ranging from public sector financial reporting and accountability, budgeting reforms, internal control mechanisms, audit quality, fiscal transparency, to the application of emerging technologies in public finance management. Each contribution reflects the authors' depth of research and commitment to the continuous improvement of public sector governance and accountability.

COLLEGE OF PUBLIC SECTOR ACCOUNTING
ANUK Public Sector Accounting and Finance Journal (APSAFJ)

THE OBJECTIVE OF THE JOURNAL

To provide a platform for researchers and academics to publish research papers and articles within the research domains of public sector accounting, public finance, public economics, environmental, oil and gas accounting, public financial management, taxation and fiscal policy, public policy and finance, macroeconomics and development, public treasury management, and other related areas..

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The standard format for content is as follows;

1. Introduction

2. Statement of Research Problem

3. Research Hypotheses or Objectives

*Review of Related Literature

4. Data Presentation & Analysis

5. Research Findings & Discussion of Findings

6. Conclusion, Recommendation & Policy Implication

7. References

8. Appendices

Or

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BIDIRECTIONAL EFFECTS OF DISRUPTIVE FINTECH INVESTMENT ON DEPOSIT MONEY BANKS' SUBSCRIPTION-BASED MODELS IN NIGERIA

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Abstract

This study investigates the bidirectional effects of disruptive FinTech investment on subscription-based models among deposit money banks in Nigeria. Adopting a quantitative cross-sectional survey design with embedded temporal lagging (2024–2025), primary data were obtained from 300 senior managers across seven purposively selected banks. Structural Equation Modeling (SEM) using SPSS v25 was employed to test the reciprocal relationships between three dimensions of disruptive FinTech investment: FinTech proxied by Investment Intensity (FII), FinTech Innovation Activity (FIA), and Investment in FinTech Infrastructure (IFI) on two subscription-based performance indicators: Subscription Performance Index (SPI) and Subscriber Retention Rate Perception (SRRP). Results indicate that all three FinTech investment dimensions in 2024 significantly enhanced subscription performance in 2025, with FII ($\beta = .17, p < .01$), FIA ($\beta = .21, p < .01$), and IFI ($\beta = .24, p < .001$) positively predicting SPI, while FII ($\beta = .14, p < .05$), FIA ($\beta = .18, p < .01$), and IFI ($\beta = .22, p < .01$) improved SRRP. Conversely, 2024 subscription performance also significantly predicted increased 2025 FinTech investment across all dimensions, confirming a dynamic, time-ordered feedback loop. The findings establish a self-reinforcing cycle of digital transformation where FinTech investment drives subscription performance, and performance success stimulates further technological investment. The study recommends that banks institutionalize performance-driven FinTech budgeting, embed innovation-performance feedback in product development, and prioritize infrastructure scalability. Regulators such as the CBN and FRC should standardize subscription KPIs, while NITDA and academic institutions should enhance capacity building for digital investment agility and cybersecurity management.

Keywords: Disruptive FinTech Investment, Subscription-Based Models, Bidirectional Causality, FinTech Infrastructure, Digital Banking, Structural Equation Modeling, Nigeria.

1.1 INTRODUCTION

In the past decade, financial technology (FinTech) has reshaped global financial services by enabling scalable, automated, and subscription-oriented business models. Globally, firms such as Stripe, Klarna, and subscription banking platforms have demonstrated how digital investments can shift revenue from transaction fees to recurring subscriptions. Recent studies highlight that FinTech adoption is accelerating firm performance and customer loyalty (Whitler, 2016; Iyengar et al., 2020). On a regional scale, in Africa, a wave of digital banking and payments innovations has disrupted traditional banking models, particularly in markets with low branch density and high mobile penetration. Nigeria, as one of Africa's leading FinTech hubs, has seen explosive growth in digital payments, neo-banks, and embedded finance, with regulatory bodies adapting to support innovation (Chambers, 2025). Yet, traditional deposit money banks in Nigeria still struggle

to fully transform their legacy systems. Given this backdrop, this study investigates the Bidirectional Effects of Disruptive FinTech Investment on Deposit Money Banks' Subscription-Based Models in Nigeria and aim to posit that disruptive FinTech investment can enhance subscription-based models by reducing legacy costs, automating operations, and improving scalability (Fuster et al., 2019). This study focuses on Access Bank PLC, Fidelity Bank PLC, First Bank of Nigeria PLC, Guaranty Trust Holding Company PLC, United Bank for Africa PLC, Zenith Bank PLC, and Wema Bank PLC.

1.2 Statement of the Problem

Despite the accelerating pace of FinTech adoption globally, the evidence on how investments in FinTech affect the viability and performance of subscription models in traditional banking remains inconclusive. Many existing studies rely on data aggregated over five

years or more, which mask short-term dynamics or reverse causality. Moreover, the most recent empirical investigations (e.g., Adewumi & Ige, 2023; Chinoda & Kapingura, 2024; Zixuan et al., 2024; Barua et al., 2025) scarcely account for bidirectional effects how improvements in subscription models might, in turn, influence further FinTech investment. As of 2025, no published work specifically examines how FinTech investment intensity, innovation activity, and infrastructure spending by Nigerian banks affect both their subscription performance and subscriber retention, while simultaneously considering the feedback effect in the reverse direction. This gap limits our understanding of the dynamic interplay underpinning successful digital transformation in Nigeria's banking sector.

1.3 Objectives of the Study

1. To examine the bidirectional effect of FinTech Investment Intensity on Subscription Performance Index.
2. To assess the bidirectional effect of FinTech Innovation Activity on Subscription Performance Index.
3. To analyze the bidirectional effect of Investment in FinTech Infrastructure on Subscription Performance Index.
4. To examine the bidirectional effect of FinTech Investment Intensity on Subscriber Retention Rate Perception.
5. To assess the bidirectional effect of FinTech Innovation Activity on Subscriber Retention Rate Perception.
6. To analyze the bidirectional effect of Investment in FinTech Infrastructure on Subscriber Retention Rate Perception.

1.4 Hypotheses of the Study

H0_{1a}: FinTech Investment Intensity has no significant bidirectional effect on Subscription Performance Index.

H0_{1b}: Subscription Performance Index has no significant bidirectional feedback effect on FinTech Investment Intensity.

H0₂: FinTech Innovation Activity has no significant bidirectional influence on Subscription Performance Index.

H0₃: Investment in FinTech Infrastructure has no significant bidirectional effect on Subscription Performance Index.

H0_{4a}: FinTech Investment Intensity has no significant bidirectional effect on Subscriber Retention Rate Perception.

H0_{4b}: Subscriber Retention Rate Perception has no significant bidirectional feedback effect on FinTech Investment Intensity.

H0₅: FinTech Innovation Activity has no significant bidirectional effect on Subscriber Retention Rate Perception.

H0₆: Investment in FinTech Infrastructure has no significant bidirectional effect on Subscriber Retention Rate Perception.

This research offers both theoretical and practical insights by modeling bidirectional dynamics between FinTech investment and subscription-based performance in Nigerian banks. It will inform bank executives, regulators, and policymakers about which investment dimensions most strongly drive subscription resilience. Understanding these mechanisms can guide resource allocation, strategy, and regulation, thereby accelerating sustainable digital banking transformation in Nigeria.

2.1 LITERATURE REVIEW

The bidirectional effects of disruptive FinTech investment on subscription-based banking models have garnered significant scholarly attention in recent years. Prior studies have conceptualized FinTech investment using proxies such as investment intensity, innovation activity, and infrastructure investment, linking them to subscription-based model performance and customer retention outcomes (Fuster et al., 2019; Ozili, 2023). These relationships are examined in the context of Nigerian deposit money banks, synthesizing conceptual, empirical, and theoretical insights into how digital transformation reshapes value delivery and customer interaction within subscription-oriented banking systems. Focusing on seven selected Deposit Money Banks (DMBs) in Nigeria.

2.1 Variables Clarified

FinTech investment intensity: refers to the proportion of financial resources committed to digital financial technologies relative to a bank's total capital or operational expenditure. It is often defined as the ratio of technology or digital expenditure to total capital expenditure (CAPEX), such as the annual spend on proprietary FinTech platforms divided by total annual CAPEX (Lee & Shin, 2018). Another approach defines it as the share of total assets devoted to FinTech-enabled services such as digital lending, payments, or online platform operations (Ozili, 2020). The first definition provides a clear metric for internal resource allocation and direct measurability but may understate investment in external collaborations or FinTech

acquisitions. Conversely, the second approach captures broader external engagement but risks conflating general digital expansion with discrete capital investment (Navaretti et al., 2018). In the Nigerian context, measuring proprietary internal expenditure better isolates the degree of disruptive investment focus, while partnership spending remains more difficult to quantify accurately (Okoye & Eze, 2022).

FinTech Innovation Activity: can be conceptualized as the number of new digital features, modules, or services launched on a bank's FinTech platform within a specific timeframe (Fuster et al., 2019). Alternatively, it may be operationalized through research and development (R&D) expenditures, innovation-specific spending in digital financial services, or the percentage of revenue derived from recently introduced FinTech products (Bunea et al., 2016). While counting the number of innovations is intuitive, it treats all innovations equally regardless of their strategic impact. The expenditure-based approach provides a more weighted measure but can suffer from lag effects or external market influences (Lee & Shin, 2018). In this study, the count-based approach is adopted for its clarity, with the possibility of integrating intensity or outcome-based indicators for robustness.

Investment in FinTech Infrastructure: denotes the scale of investment in physical and digital systems such as hardware, cloud computing, data centers, and network upgrades that support financial technology operations (Gomber et al., 2018). It can be measured either quantitatively through capital spending or qualitatively via digital readiness or technology maturity indices (Zavolokina et al., 2016). The quantitative definition is objective and expenditure-based but may face comparability issues due to differing cost structures across banks. Conversely, maturity indices capture qualitative dimensions of readiness but depend heavily on perception, introducing bias (Ozili, 2023). In Nigeria's heterogeneous banking landscape, combining both objective expenditure data and subjective readiness scales yields a more valid assessment of infrastructure investment.

The Subscription Performance Index (SPI): captures a bank's success in leveraging subscription-based services to drive growth and profitability. SPI can be measured as the aggregated mean of managerial perceptions regarding subscription revenue growth, profitability, and overall contribution to total revenue (Fuster et al., 2019). Alternatively, SPI can rely on objective indicators such as the ratio of subscription income to total revenue or the growth rate of

subscription-based earnings (McKinsey & Company, 2021). While perception-based indices are feasible and practical in contexts with limited financial disclosures, they may introduce managerial bias. Objective measures are more reliable but may be unavailable or inconsistent across institutions (Okoye & Eze, 2022). Thus, in Nigeria, a hybrid approach combining perception and verifiable revenue proxies enhances validity.

Subscriber Retention Rate Perception (SRRP): represents management's perceived success in retaining customers within subscription-based financial services. It may be measured via Likert-scale survey items evaluating agreement with statements on customer churn reduction and improved retention over the past year (Kim et al., 2020). Alternatively, SRRP can be derived from actual churn or retention metrics if internal data are available (Homburg et al., 2017). The perception-based measure is cost-effective and operationally feasible but can be influenced by bias or optimism effects. Objective retention data, though more accurate, are often confidential or inconsistently tracked (Ozili, 2023). Given data limitations in Nigerian banks, perceptual measures are most practical, with triangulation through available internal data recommended for improved reliability.

2.2 FinTech Investment Determinants on SPI and SRRP

FinTech Investment Intensity and Subscription Performance Index (SPI)

High FinTech investment intensity is theorized to directly enhance subscription performance by reducing legacy system costs, increasing operational scalability, and enabling frequent updates that sustain subscription growth (Feyen et al., 2021). Greater digital capital allocation can enhance service delivery and performance outcomes through automation and cost efficiencies (Lee & Shin, 2018). However, evidence from digital transformation studies suggests that excessive or misdirected investment may yield diminishing marginal returns (Navaretti et al., 2018). Because this thesis postulates bidirectional relationships, it is also plausible that improved Subscription Performance Index (SPI) outcomes could encourage further FinTech investment. That is, banks experiencing strong subscription performance may reinvest profits into additional digital innovations, producing a reinforcing feedback loop (Fuster et al., 2019; Ozili, 2023). Nonetheless, empirical studies confirming such feedback effects remain limited,

especially within the context of banking subscription models in developing economies.

FinTech Innovation Activity and Subscription Performance Index (SPI)

Innovation activity represented by the introduction of new features, modules, or FinTech-enabled products enhances the value proposition of subscription-based offerings and can strengthen customer engagement (Vaska et al., 2021). Consistent with business model innovation theory, continuous FinTech innovation differentiates a bank's digital ecosystem and drives long-term performance (ResearchGate, 2023). Yet, innovation efforts that lack supporting infrastructure or fail to meet market readiness may not translate into improved subscription outcomes (Gomber et al., 2018). From a bidirectional perspective, high SPI performance may also stimulate further innovation activity, as banks that experience success in their digital subscriptions often allocate more resources to research and development (Bunea et al., 2016). Such performance-driven reinvestment can reinforce competitive advantage but may also risk over-innovation if customer needs are not clearly understood (Homburg et al., 2017).

Investment in FinTech Infrastructure and Subscription Performance Index (SPI)

Robust FinTech infrastructure comprising hardware, cloud systems, data centers, and networks provides the backbone for scalable and resilient subscription platforms (Feyen et al., 2021). The maturity of digital infrastructure enhances transaction speed, system reliability, and automation capacity, directly supporting the performance of subscription-based offerings (Gomber et al., 2018). However, such investments involve large sunk costs and long gestation periods, which can deter consistent funding among Nigerian banks (Okoye & Eze, 2022). In the reverse direction, banks experiencing high SPI may also justify and fund further infrastructure upgrades, viewing such reinvestments as strategic enablers for sustaining subscription performance and competitive advantage (McKinsey & Company, 2021).

FinTech Investment Intensity and Subscriber Retention Rate Perception (SRRP)

Sustained FinTech investment intensity supports customer retention by enabling continuous service enhancement, faster issue resolution, and feature updates. Because subscription-based banking models depend on customer loyalty and low churn, higher

FinTech spending can promote sustained engagement and retention (Kim et al., 2020). However, overinvestment in backend systems without a proportional focus on customer experience may not yield expected improvements in retention (Homburg et al., 2017). The relationship may also operate in the opposite direction: banks perceiving high Subscriber Retention Rate Perception (SRRP) may view stable subscriber bases as a signal of success, thus justifying or expanding FinTech investments to preserve customer satisfaction and further lock in loyalty (Ozili, 2023).

FinTech Innovation Activity and Subscriber Retention Rate Perception (SRRP)

Innovation activity fosters subscriber retention by providing continuous value through new or improved digital experiences. Incremental innovations such as personalized dashboards, real-time notifications, or AI-driven insights can strengthen customer loyalty and prevent "subscription fatigue" (Vaska et al., 2021). However, poorly implemented innovations or disruptive interface changes can produce user dissatisfaction and increase churn (Fuster et al., 2019). From a bidirectional standpoint, banks with high SRRP are often more confident in pursuing new innovations, leveraging positive customer feedback to guide and justify additional product development (Kim et al., 2020). This cyclical relationship reinforces both innovation and loyalty, aligning with theories of continuous service improvement.

Investment in FinTech Infrastructure and Subscriber Retention Rate Perception (SRRP)

FinTech infrastructure quality manifested through uptime reliability, system responsiveness, and data security directly affects user satisfaction and retention (Gomber et al., 2018). High-quality infrastructure ensures uninterrupted service delivery, a critical determinant of retention in subscription environments. Conversely, weak or outdated infrastructure often leads to latency, errors, and downtime that drive customer churn (Feyen et al., 2021). In feedback terms, a high SRRP can motivate banks to reinvest in infrastructure as part of efforts to sustain service reliability and customer trust (McKinsey & Company, 2021). This reinforces a dynamic investment-retention loop consistent with digital capability-building frameworks (Lee & Shin, 2018).

2.2 Empirical Review

2.2.1 FinTech Investment Intensity and Subscription Performance Index (SPI)

Feyen et al. (2021) conducted a study on *FinTech investment and digital transformation in financial services* using cross-country data from 2015–2020. Their findings revealed that higher FinTech investment intensity measured as digital expenditure relative to total capital expenditure significantly improved revenue performance and customer onboarding efficiency, key components of subscription performance. Similarly, Lee and Shin (2018) analyzed 60 global financial institutions and found that greater technological spending correlated positively with scalability and subscription-based profitability. In Nigeria, Okoye and Eze (2022) examined digital financial transformation and performance of deposit money banks between 2012–2020. They observed that FinTech investment intensity significantly enhanced digital revenue streams and subscription-led income growth. However, they cautioned that overinvestment could lead to resource rigidity and diminishing returns, suggesting a curvilinear relationship.

2.2.2 FinTech Innovation Activity and Subscription Performance Index (SPI)

Vaska et al. (2021) conducted a systematic review of 88 studies on digital transformation and business model performance across European financial technology firms. Their results indicated that innovation activity proxied by the number of new digital features and services had a strong positive effect on performance, especially in subscription-based service environments. Similarly, Fuster et al. (2019) analyzed data from major U.S. banks adopting FinTech platforms between 2010–2018 and found that new FinTech innovations significantly increased subscription user growth and service uptake. In Africa, Ozili (2023) found that innovation activity among Nigerian and Kenyan banks led to improved customer subscription performance by enhancing convenience and trust in digital transactions. However, Bunea et al. (2016) noted that while innovation enhances service performance, the absence of supporting infrastructure and readiness can dampen its overall impact. This suggests that innovation activity is most effective when complemented by adequate technological infrastructure.

2.2.3 Investment in FinTech Infrastructure and Subscription Performance Index (SPI)

Gomber et al. (2018) studied FinTech infrastructure readiness and service performance using firm-level data from European banks and found a significant positive association between digital infrastructure maturity and performance indices such as service uptime, digital scalability, and subscription profitability. Feyen et al. (2021) reinforced these findings, emphasizing that core infrastructure investment lowers transaction costs and enhances subscription service efficiency. In the Nigerian context, Okoye and Eze (2022) reported that digital infrastructure investments, including data centers and cloud platforms, improved subscription-related income and transaction reliability among deposit money banks. Yet, their study also highlighted challenges of high capital intensity and delayed returns on infrastructure projects, which can reduce short-term financial benefits.

2.2.4 FinTech Investment Intensity and Subscriber Retention Rate Perception (SRRP)

Kim et al. (2020) examined customer retention in FinTech services using a survey of 1,200 digital banking users in South Korea. They found that higher FinTech investment intensity indirectly enhanced customer retention through improved service quality and continuous feature upgrades. Similarly, Homburg et al. (2017) observed that sustained investment in digital systems reduced customer churn by improving responsiveness and personalization. In Nigeria, Ozili (2023) found that banks with consistent FinTech investments achieved higher customer retention, as such investments allowed real-time service access, instant payments, and seamless subscription renewals. However, the study also revealed that retention gains were lower when technological investments were not user-centered or poorly communicated to customers.

2.2.5 FinTech Innovation Activity and Subscriber Retention Rate Perception (SRRP)

Fuster et al. (2019) established a positive and significant link between FinTech innovation activity and customer retention in digital subscription banking. They showed that continuous feature innovation creates switching barriers and enhances customer loyalty. Likewise, Kim et al. (2020) emphasized that innovation-driven convenience and personalization strongly influence subscriber retention intentions. Vaska et al. (2021) also observed that firms engaging in frequent innovation cycles tend to experience sustained retention due to improved customer

satisfaction. However, Homburg et al. (2017) warned that excessive or disruptive innovation such as frequent interface changes can reduce retention if customers perceive instability in the digital experience.

2.2.6 FinTech Infrastructure and Subscriber Retention Rate Perception (SRRP)

Gomber et al. (2018) demonstrated that stable FinTech infrastructure measured through system uptime, response speed, and security protocols has a strong positive impact on customer retention. Inadequate infrastructure, on the other hand, leads to service downtimes that significantly increase churn rates. Feyen et al. (2021) further confirmed that investment in digital infrastructure enables consistent service quality, fostering long-term retention. In the Nigerian setting, Okoye and Eze (2022) found that digital infrastructure modernization was the single most influential predictor of perceived subscriber retention, as improved system reliability directly enhanced trust and usage continuity.

2.3 Theoretical Review

Disruptive Innovation Theory

Disruptive innovation refers to processes by which new entrants introduce products or services that initially target underserved or low-end segments but gradually evolve to displace incumbent offerings. The theory holds that incumbents often fail when they neglect emerging disruptive technologies due to focus on sustaining innovations (Christensen et al., 2015). FinTech investment by banks can be viewed as an internal effort to adopt disruptive innovations transforming legacy banking systems into modular, scalable subscription-based platforms. According to disruption logic, these investments should enable banks to create new value propositions (subscription models) that eventually challenge the traditional transaction-based model. The strength of this theory lies in its focus on how disruptive technologies emerge and alter industry structures, aligning well with your variable of FinTech investment.

Strengths / Weaknesses concerning variables:

Strengths: Disruptive innovation theory nicely explains why banks need to invest in innovation activity and infrastructure: the theory posits that disruptive entrants succeed by leveraging new architectures and cost structures. It also accommodates the idea of dynamic feedback once subscription models

succeed, incumbents may reinvest in new technological cycles.

Weaknesses: The theory was originally conceptualized for entrant incumbents, not incumbent firms attempting self-disruption. It underemphasizes managerial choice and internal capabilities (Lile, 2024). When applied to banks, it may overstate technology as a driver and understate regulatory, cultural, or institutional constraints in Nigeria. Also, the theory gives less direct guidance on retention dynamics or subscription-specific metrics.

3.0 METHODOLOGY

Research Design

This study adopts a quantitative cross-sectional survey design enhanced with lagged temporal anchoring to examine the bidirectional relationships between disruptive FinTech investment and subscription-based models in Nigerian deposit money banks. To ensure methodological consistency and compatibility with Structural Equation Modeling (SEM), all constructs including FinTech investment dimensions are operationalized as multi-item, 5-point Likert-type latent variables, anchored to distinct time periods (2024 vs. 2025). Specifically, the survey embeds retrospective time cues (as “In 2024...” and “So far in 2025”) to enable estimation of cross-lagged predictive paths:

FinTech investment (2023) → Subscription performance (2024)

Subscription performance (2023) → FinTech investment (2024)

This design provides a valid approximation of reciprocal, time-ordered relationships within a single data collection wave (Bharadwaj et al., 2013; Fügener et al., 2021). Data were analyzed using SEM in SPSS (v25) to simultaneously test bidirectional paths while accounting for measurement error through confirmatory factor analysis (CFA). The approach is appropriate for capturing managerial perceptions of strategic investment and performance dynamics in an emerging market context where objective financial disclosures on FinTech are often limited or non-standardized.

Population, Scope, and Sampling Frame

The target population comprises senior managers and executives involved in FinTech strategy, digital innovation, or subscription-based models product management across seven purposively selected Deposit Money Banks (DMBs) in Nigeria: Access Bank PLC, Fidelity Bank PLC, First Bank of Nigeria PLC, Guaranty Trust Holding Company PLC, United Bank for Africa PLC, Zenith Bank PLC, and Wema Bank PLC. These banks were selected based on visible FinTech engagement and subscription-based models. The scope covers the bidirectional relationships among five latent constructs: FinTech Investment Intensity (FII), FinTech Innovation Activity (FIA), Investment in FinTech Infrastructure (IFI), Subscription Performance Index (SPI), Subscriber Retention Rate Perception (SRRP). An estimated 100 eligible respondents per bank (total population = 700) were identified from Finance, Product/IT, and Digital Banking units. Using Yamane's (1967) formula at a 5% margin of error, the minimum required sample was 255. The study targeted 300 respondents to offset non-response; 300 complete responses were obtained and used for analysis.

Using Yamane's (1967) formula for finite populations at a 5% margin of error:

$$n = \frac{N}{1 + N(e^2)} = \frac{700}{1 + 700(0.05)^2} = \frac{700}{1 + 17.5} = 254.5 \approx 255$$

Construct Operationalization (All Latent, Multi-Item, 5-Point Likert). All variables were measured using 5-point Likert scales (1 = Strongly Disagree, 5 = Strongly Agree), with items anchored to specific years. Each construct comprises 2–3 reflective indicators, ensuring reliability and validity. Each construct is measured separately for 2024 and 2025, yielding 10 latent variables in the full SEM model (FII_2024, SPI_2025).

Model Specification

The study tests cross-lagged structural models using time-anchored variables:

Model Set A: FinTech Investment (2024) on Subscription-based models(2025)

Table 1. Reliability and Validity Results (Pilot Test, N = 25)

Construct	No. of Items	Cronbach's Alpha	Composite Reliability (CR)	Average Variance Extracted (AVE)
FinTech Investment Intensity (FII)	3	0.85	0.87	0.65
FinTech Innovation Activity	3	0.83	0.86	0.63

$$\text{Model 1: } SPI_{2025} = \beta_0 + \beta_1(FII_{2024}) + \beta_2(FIA_{2024}) + \beta_3(IFI_{2024}) + \varepsilon_1$$

$$\text{Model 2: } SRRP_{2025} = \beta_0 + \beta_1(FII_{2024}) + \beta_2(FIA_{2024}) + \beta_3(IFI_{2024}) + \varepsilon_2$$

Model Set B: Subscription-based models (2024) on FinTech Investment (2025)

$$\text{Model 3: } FII_{2025} = \beta_0 + \beta_1(SPI_{2024}) + \beta_2(SRRP_{2024}) + \varepsilon_3$$

$$\text{Model 4: } FIA_{2025} = \beta_0 + \beta_1(SPI_{2024}) + \beta_2(SRRP_{2024}) + \varepsilon_4$$

$$\text{Model 5: } IFI_{2025} = \beta_0 + \beta_1(SPI_{2024}) + \beta_2(SRRP_{2024}) + \varepsilon_5$$

Where:

SPI = Subscription Performance Index

SRRP = Subscriber Retention Rate Perception

FII = FinTech Investment Intensity

FIA = FinTech Innovation Activity

IFI = Investment in FinTech Infrastructure

β_0 = Intercept; β_1 – β_3 = Standardized path coefficients; ε = Error term

Note: All variables are time-specific (T–1 or T), enabling identification of reciprocal paths in a non-recursive SEM framework.

Validity and Reliability of Instrument

Face and content validity were established through expert review by two academics (University of Nigeria, Nsukka) and two industry practitioners (Lagos Business School), ensuring alignment with disruptive innovation theory and Nigerian banking context. A pilot test (N = 25) confirmed internal consistency. Cronbach's Alpha and Composite Reliability (CR) exceeded thresholds:

Construct	No. of Items	Cronbach's Alpha	Composite Reliability (CR)	Average Variance Extracted (AVE)
(FIA)				
Investment in FinTech Infrastructure (IFI)	3	0.82	0.85	0.61
Subscription Performance Index (SPI)	3	0.88	0.90	0.71
Subscriber Retention Rate Perception (SRRP)	2	0.81	0.84	0.72

Estimation Technique: Descriptive Statistics, Normality Tests, Measurement Model – Reliability and Validity (CFA Results), Model Fit Indices (Full Cross-Lagged SEM), and Structural Model Results (Standardized Path Coefficients).

A priori expectations: Higher FinTech investment in 2024 across all three dimensions (FII, FIA, and IFI) will positively predict subscription-based performance in 2025 (i.e., higher SPI and SRRP). Stronger subscription-based performance in 2024 (SPI and SRRP) will positively predict increased FinTech investment in 2025 across all three dimensions (FII, FIA, and IFI).

4.0 RESULTS AND DISCUSSIONS

4.1 Data Analyses

Table 4.1.1: Descriptive Statistics

Table 1: Descriptive Statistics (N = 300)

(Based on latent construct / factor scores)

Variable	Year	Mean	SD	Min	Max
FinTech Investment Intensity (FII)	2024	3.42	0.81	1.00	5.00
	2025	3.68	0.79	1.20	5.00
FinTech Innovation Activity (FIA)	2024	3.25	0.85	1.00	4.90
	2025	3.51	0.83	1.10	5.00
FinTech Infrastructure Investment (IFI)	2024	3.30	0.87	1.00	4.95
	2025	3.55	0.84	1.15	5.00
Subscription Performance Index (SPI)	2024	3.38	0.76	1.20	4.90
	2025	3.72	0.72	1.40	5.00
Subscriber Retention Perception (SRRP)	2024	3.15	0.82	1.00	4.80
	2025	3.48	0.79	1.20	5.00

Source: SPSS v25 Descriptive Output (2025).

Interpretation: Descriptive results show consistent improvement across all FinTech and performance constructs from 2024 to 2025. Mean increases in FII (3.42→3.68), FIA (3.25→3.51), and IFI (3.30→3.55) suggest progressive FinTech investment intensity and capability. Concurrent rises in SPI (3.38→3.72) and SRRP (3.15→3.48) indicate enhanced subscription performance and customer retention perception, implying a positive dynamic between FinTech engagement and subscription-based outcomes.

Table 2: Normality Tests (Based on Item-Level Data)

Variable	Shapiro–Wilk (W)	p-value	Skewness	Kurtosis
FII_2024	0.976	0.001	-0.32	-0.41
FIA_2024	0.971	<0.001	-0.28	-0.35
IFI_2024	0.979	0.003	-0.35	-0.48
SPI_2024	0.983	0.012	-0.40	-0.22
SRRP_2024	0.980	0.005	-0.37	-0.29
FII_2025	0.974	<0.001	-0.30	-0.39
FIA_2025	0.969	<0.001	-0.25	-0.31
IFI_2025	0.977	0.002	-0.33	-0.44
SPI_2025	0.981	0.007	-0.42	-0.20
SRRP_2025	0.978	0.004	-0.39	-0.26

Source: SPSS v25 (2025).

Normality assessment generated using : All $p < 0.05 \rightarrow$ non-normal, but $|\text{skewness}| < 0.5$ and $|\text{kurtosis}| < 1 \rightarrow$ acceptable for SEM (Hair et al., 2019).

Interpretation: Although the Shapiro–Wilk tests show significant p-values ($p < 0.05$), indicating non-normality, the skewness and kurtosis values are within acceptable ranges ($|\text{skewness}| < 0.5$; $|\text{kurtosis}| < 1$). This confirms that the data approximate normal distribution closely enough for SEM estimation, which is robust to mild deviations from normality.

Table 3: Measurement Model – Reliability and Validity (CFA Results)

Construct	No. of Items	Cronbach's α	CR	AVE
FII	3	0.85	0.87	0.65
FIA	3	0.83	0.86	0.63
IFI	3	0.82	0.85	0.61
SPI	3	0.88	0.90	0.71
SRRP	2	0.81	0.84	0.72

Source: SPSS v25 (2025).

(CFA) performed

Interpretation: All constructs exhibit excellent internal consistency and convergent validity, with Cronbach's $\alpha > 0.80$, CR > 0.84 , and AVE > 0.61 . The Fornell–Larcker criterion further confirms discriminant validity, ensuring that each latent variable is statistically distinct from the others. Hence, the measurement model is psychometrically sound and suitable for subsequent SEM analysis.

Table 4: Model Fit Indices (Full Cross-Lagged SEM)

χ^2	df	χ^2/df	CFI	TLI	RMSEA	[90% CI]
412.6	210	1.97	0.948	0.939	0.056	[0.048–0.064]

Source: SPSS v25 (2025).

Interpretation: The full cross-lagged SEM model demonstrates excellent overall fit. All indices satisfy recommended thresholds: CFI (0.948) and TLI (0.939) exceed 0.90; RMSEA (0.056) is below 0.08; and χ^2/df (1.97) is below 3. This indicates that the hypothesized bidirectional structural relationships align well with the observed data.

Table 5: Structural Model Results (Standardized Path Coefficients)

Model 1: FinTech Investment (2024) → Subscription Performance (2025)

Path	β	SE	t	p	Sig.
FII ₂₀₂₄ → SPI ₂₀₂₅	0.17	0.06	2.83	0.005	✓
FIA ₂₀₂₄ → SPI ₂₀₂₅	0.21	0.07	3.00	0.003	✓
IFI ₂₀₂₄ → SPI ₂₀₂₅	0.24	0.07	3.43	<0.001	✓

Model 2: FinTech Investment (2024) → Subscriber Retention (2025)

Path	β	SE	t	p	Sig.
FII ₂₀₂₄ → SRRP ₂₀₂₅	0.14	0.06	2.33	0.020	✓
FIA ₂₀₂₄ → SRRP ₂₀₂₅	0.18	0.07	2.57	0.010	✓
IFI ₂₀₂₄ → SRRP ₂₀₂₅	0.22	0.07	3.14	0.002	✓

Model 3: Subscription Outcomes (2024) → FinTech Investment Intensity (2025)

Path	β	SE	t	p	Sig.
SPI ₂₀₂₄ → FII ₂₀₂₅	0.15	0.06	2.50	0.012	✓
SRRP ₂₀₂₄ → FII ₂₀₂₅	0.13	0.06	2.17	0.030	✓

Model 4: Subscription Outcomes (2024) → FinTech Innovation Activity (2025)

Path	β	SE	t	p	Sig.
SPI ₂₀₂₄ → FIA ₂₀₂₅	0.19	0.07	2.71	0.007	✓
SRRP ₂₀₂₄ → FIA ₂₀₂₅	0.16	0.07	2.29	0.022	✓

Model 5: Subscription Outcomes (2024) → FinTech Infrastructure Investment (2025)

Path	β	SE	t	p	Sig.
SPI ₂₀₂₄ → IFI ₂₀₂₅	0.18	0.07	2.57	0.010	✓
SRRP ₂₀₂₄ → IFI ₂₀₂₅	0.15	0.07	2.14	0.032	✓

Source: SPSS v25 (2025).

Cross-lagged structural paths estimated

Interpretation: All paths are statistically significant ($p \leq 0.05$), establishing strong bidirectional relationships between FinTech investments and subscription-based outcomes across time. Specifically, FinTech investment intensity, innovation activity, and infrastructure in 2024 significantly predict 2025 subscription performance (SPI) and retention perception (SRRP). Conversely, higher subscription performance and retention in 2024 significantly predict increased FinTech investments in 2025

5.1 CONCLUSIONS

This study confirms that disruptive FinTech investment and subscription-based performance in Nigerian deposit money banks are linked through a dynamic, bidirectional feedback loop. Using cross-sectional data (2024–2025) and SEM analysis, it reveals that FinTech Investment Intensity (FII), FinTech Innovation Activity (FIA), and Investment in FinTech Infrastructure (IFI) not only enhance subscription outcomes but are themselves strengthened by prior performance, forming a self-sustaining cycle of digital transformation and strategic reinvestment.

Recommendations:

FinTech Investment Intensity (FII): Banks should institutionalize quarterly performance–investment review cycles where SPI and SRRP metrics directly inform annual FinTech budgeting. This ensures that investment intensity aligns with measurable outcomes in profitability and retention. Chief Digital Officers and CFOs should co-lead these reviews to maintain strategic coherence between financial performance, reinvestment priorities, and digital transformation objectives across business units.

FinTech Innovation Activity (FIA): To sustain innovation responsiveness, banks should embed performance feedback into product development. Agile sprints should be triggered by quarterly SPI and SRRP reviews, linking innovation output to customer retention and subscription profitability. Product and Digital Banking divisions must develop real-time dashboards translating performance data into prioritized feature rollouts, ensuring that innovation continuously reflects evolving market performance indicators.

Investment in FinTech Infrastructure (IFI): Given its dual influence on performance and reinvestment, banks should dedicate at least 20% of annual FinTech budgets to scalable, modular, cloud-based infrastructure. These systems must include real-time analytics for adjusting capacity based on subscriber growth. Implementation should be managed by Technology and Innovation Units, adhering to NITDA cybersecurity standards to ensure resilience, scalability, and sustained digital competitiveness.

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